RESEARCH MONOGRAPH

PROJECT HANNIBAL

Evidence, Equity, Recoverability and Human Agency in Complex Systems

A research monograph on informational equity, governance chokepoints, evidence-gated intelligence and recoverability.

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Publication notice

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To people who have felt the consequences of a decision without being shown the process that produced it; to volunteers who hold communities together; and to those who keep asking for evidence when authority asks only for trust.

Contents

Acknowledgments . . . 7

Author's foreword . . . 7

How to read this monograph . . . 8

Executive summary . . . 8

Acronyms . . . 9

Part I · The Argument . . . 10

1.1 Standpoint and disclosure . . . 10

1.2 The crisis generation and the limits of analogy . . . 10

1.3 Abundance without comprehension . . . 11

1.4 Informational equity and human agency . . . 11

1.5 The thesis and how it could fail . . . 11

1.6 Research questions, scope and method . . . 12

1.7 Position relative to existing practice . . . 12

1.8 Contributions and their evidential status . . . 13

Part II · Berlin and Belgium . . . 14

2.1 Berlin as a natural experiment . . . 14

2.2 What Berlin does not show . . . 14

2.3 Belgium: three regions, one polity . . . 14

2.4 Divergent adaptation . . . 15

2.5 Brussels: decision density and the commuter effect . . . 16

2.6 Language, plurality and cohesion . . . 16

2.7 From cases to requirements . . . 17

Part III · Technical Architecture . . . 18

3.1 Design principles . . . 18

3.2 From feeds to events . . . 18

3.3 The evidence record . . . 19

3.4 Scoring evidence . . . 19

3.5 Aggregate confidence . . . 20

3.6 Composite friction and the chokepoint multiplier . . . 20

3.7 Signals and their limits . . . 22

3.8 Arbitration, the confidence gate and human review . . . 22

3.9 The language model and the interface . . . 23

3.10 Audit, security and failure semantics . . . 23
3.11 Architecture synthesis . . . 24

Part IV · Recoverability, Governance and Validation . . . 25
4.1 Recoverability defined . . . 25
4.2 Governance nodes . . . 25
4.3 Equity evaluation . . . 25
4.4 Ethical boundaries . . . 26
4.5 Validation design . . . 26
4.6 Red teaming and governance of Hannibal . . . 27
4.7 Objections and replies . . . 27
4.8 Limitations and research agenda . . . 28

Afterword . . . 29

Appendices . . . 30
Appendix A · Core equations . . . 30
Appendix B · Data schemas . . . 30
Appendix C · Acceptance tests . . . 31
Appendix D · Author and CV note . . . 32
Appendix E · Ledger of central claims . . . 32
Appendix F · Reporting format for validation results . . . 34

References . . . 35

Figures

Figure 1 GDP per head in purchasing power standards, 2018, as a percentage of the EU average, by Belgian province and the Brussels-Capital Region 15
Figure 2 The Hannibal evidence pipeline. Every stage writes to the audit ledger. 18
Figure 3 Friction index F against base stress B for three values of the chokepoint measure h 21
Figure 4 The confidence gate. Mandatory triggers are checked before the score threshold. 23

Tables

Table 1	Contributions and the status of the support for each	13
Table 2	Employment rate, ages 20–64, 2023 (%)	15
Table 3	Traceability from cases to design requirements	17
Table 4	Pipeline stages from collection to event	19
Table 5	Source classes	19
Table 6	Aggregate confidence: worked cases (illustrative)	20
Table 7	Friction variables	21
Table 8	Friction index F for selected base scores (illustrative)	22
Table 9	Signals used as inputs and what they cannot establish	22
Table 10	Failure states and how they must not be displayed	24
Table 11	Recoverability dimensions	25
Table 12	Estimands, baselines and what would count against each claim	26
Table 13	Acceptance tests and proposed verification	31
Table 14	Ledger of central claims	33
Table 15	Aggregate tables for a validation report	34

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Author's foreword

I did not begin Project Hannibal because I wanted a machine to sound certain. I began it because too many people are expected to live with decisions they cannot inspect. I have worked across cultures, languages and institutions, including customer support, technical troubleshooting, payments, fraud-related work, healthcare support and volunteer service. Those environments taught me that the quality of an answer depends on whether it can be traced, challenged and corrected.

I came of age amid the after-effects of the global financial crisis, the euro-area crisis, the COVID-19 pandemic, inflation, supply-chain disruption and geopolitical shocks. I also grew up in a European historical landscape permanently marked by the Holocaust: a warning that bureaucracy, obedience and technical competence are not substitutes for moral responsibility. These events are not interchangeable. They do, however, reveal why institutions must remain answerable to evidence and human consequence.

This monograph argues that understandable information is a dimension of equity. Hannibal is my attempt to make complex systems more legible without pretending that uncertainty can be abolished.

How to read this monograph

Part I develops the argument: why understandable evidence is a condition of human agency, where the project sits relative to existing practice, and what would count as evidence against it. Part II studies Berlin, Flanders, Wallonia and Brussels and converts what they show into design requirements. Part III defines the architecture, its equations and their properties. Part IV treats recoverability, ethics, validation and the strongest objections. The appendices preserve equations, schemas, acceptance tests, a ledger of claims and their evidential status, and the reporting format for validation results.

Because the monograph's subject is the difference between evidence and assertion, it applies that distinction to itself. Statements are tagged by evidential status wherever the difference matters, and Appendix E lists the central claims in one place.

> ***S** sourced: supported by a cited source. **D** definition: a stipulated meaning. **N** normative premise: a value commitment, not a finding. **P** prototype rule: a design choice that has not been empirically calibrated. **R** derived result: follows from the monograph's own equations and can be reproduced by hand. **H** hypothesis: stated and testable, but untested in this text.*

Executive summary

Project Hannibal is an evidence-gated decision-support architecture whose primary objective is intelligibility rather than autonomous prediction. It collects heterogeneous public signals, normalizes them into provenance-bearing records, distinguishes signal from evidence, exposes disagreement, calculates bounded indicators and routes weakly supported claims to human review.

The monograph's thesis is that human agency depends on equitable access to understandable evidence about the systems shaping everyday life. Berlin illustrates the economic consequences of broken and restored connectivity. Belgium illustrates divergence within a connected polity, and shows how a headline indicator for its capital region reflects commuting as much as residents' welfare. Together with Brussels' concentration of decision-making, these cases motivate the question that conventional recovery statistics often omit: recovered for whom?

The technical proposal includes evidence records, source classes, freshness and contradiction handling, three-way arbitration, confidence gates, audit trails, recoverability dimensions and an explicit separation between model-generated narrative and factual evidence. The monograph also examines its own prototype rules and reports three properties that follow from them: the friction index saturates before the chokepoint multiplier can act on the highest-stress nodes; the aggregate confidence score cannot by itself block an entirely unsupported critical premise; and the ratio form of the thesis has defects that limit it to a directional statement.

Validation is specified rather than claimed. Chapter 4.5 defines estimands, baselines, safeguards against leakage and a reporting format, and Chapter 4.7 sets out the strongest objections with what would count as their vindication.

Acronyms

AI	artificial intelligence
DART	Deep-ocean Assessment and Reporting of Tsunamis
ECE	expected calibration error
EU	European Union
ICD	Intelligence Community Directive
NATO	North Atlantic Treaty Organization
OSINT	open-source intelligence
PPS	purchasing power standards
SEC	U.S. Securities and Exchange Commission

PART I

The Argument

1.1 Standpoint and disclosure

Project Hannibal emerged from a multilingual Belgian life and from work in customer support, technical troubleshooting, payments, fraud-related operations, healthcare technology and volunteer service, including at Bouckenborgh Park. That work repeatedly placed the author between institutional procedure and individual consequence, and it is the origin of three design requirements: that an answer be traceable to its sources, that a person affected by it be able to challenge it, and that the record of the challenge survive.

Biography is disclosed as a source of questions, not as authority for answers. Personal experience cannot validate a model; it can identify values and failure modes that a research program must then test. The author's stated commitment to equity is treated the same way. It changes what the system must optimize, which is not only analytical accuracy but accessibility, contestability and the fair distribution of interpretive capacity, and it is a normative premise **[N]**, not a finding.

Belgium's Dutch-, French- and German-speaking communities, and Brussels' international character, are a working example of why one narrative cannot serve every reader. They motivate an architecture that preserves disagreement instead of forcing every signal into one answer. Plurality is not romanticized: multilingual settings can also reproduce exclusion, unequal access and administrative opacity.

1.2 The crisis generation and the limits of analogy

The author's formative period included the after-effects of the 2008 financial crisis, the euro-area crisis, the pandemic, and the inflation and supply disruption that followed. These shocks showed that household experience is produced by interacting financial, health, logistics, energy and governance systems. They differ in cause, scale and moral meaning and are not collapsed here into one narrative.

Two lessons carry into the design. First, aggregate recovery can coexist with insecurity, debt, precarious work and unequal exposure to inflation, so two people living under the same macroeconomic indicator may experience opposite realities. Macroeconomic indicators remain valuable; they are insufficient alone, and "recovered for whom?" belongs at the center of the model. Second, the pandemic made models, uncertainty and institutional trust part of everyday life. Guidance changed as evidence changed, while citizens struggled to tell revision from contradiction. Hannibal's answer is versioned evidence and visible uncertainty: what was known at a time, why a conclusion changed and which assumptions were revised. This monograph does not adjudicate medical controversies and does not substitute for public-health expertise.

European civic reasoning cannot responsibly ignore the Holocaust. It demonstrates that administrative capacity, data collection and rule-following can be turned toward catastrophic injustice when institutions abandon human dignity and accountability. The lesson for an AI-assisted system is modest but non-negotiable: technical efficiency is never sufficient moral justification, so constraints, documentation, refusal and human responsibility must be built into workflows. The Holocaust is not used as a rhetorical analogy for ordinary disputes, and the monograph claims no equivalence between AI errors and genocide.

1.3 Abundance without comprehension

Digital systems produce an unprecedented quantity of public information. Yet volume can obscure origin, duplicate claims and reward speed over verification. Hannibal therefore treats collection as the beginning of analysis, not its conclusion: record structure and evidence quality matter more than feed count. No interface can eliminate propaganda, manipulation or cognitive bias.

The guiding analogy is an economic hearing aid. A hearing aid does not invent sound; it amplifies selected signals so a person can interpret them. Hannibal adopts that assistive framing for economic and infrastructure information. The user remains the decision-maker, and the system should amplify provenance, consequence and uncertainty rather than confidence theatre. The analogy has limits and is not a medical claim.

1.4 Informational equity and human agency

Informational equity is the fair capacity to access, interpret and contest consequential information, including its language, format, timing, the expertise it presumes and the responsiveness of the institutions behind it **[D]**. It extends Amartya Sen's capability orientation (Sen, 1999): information matters when it expands what people can actually understand and do. Meadows (2008) makes a related point from systems thinking, treating the structure of information flows, meaning who has access to what and when, as a lever on system behavior.

Agency, in this sense, is more than choice among options. It requires enough visibility to understand constraints, consequences and routes of challenge, and the system's value is measured partly by whether a user can explain why a conclusion was reached and what would change it. Agency does not imply that every complex system can be mastered by one person.

Power should therefore be explainable. Institutions often see individuals in high resolution while individuals see institutions only through outcomes, an asymmetry that can be administratively efficient and civically corrosive. Evidence bundles, reason codes and appeal paths can narrow it without exposing protected data or compromising legitimate security. Transparency has limits where privacy, safety, legal privilege or system integrity would be harmed, and where those limits apply they should be stated. Informational equity currently has dimensions rather than a validated index, and the concept needs empirical measures.

Values become meaningful only when translated into requirements. Equity becomes accessible explanations, accountability becomes immutable audit references, and humility becomes confidence gates. That translation links biography to technical design without claiming that code can settle moral questions, and requirements should be evaluated with affected users, not inferred solely by developers.

1.5 The thesis and how it could fail

Thesis [N]. Human agency depends on equitable access to understandable evidence about the systems that shape social, economic and institutional life. The thesis does not claim that information alone solves inequality. It claims that opacity compounds inequality by restricting the capacity to interpret and contest decisions.

The organizing relation is directional: agency rises with evidence, transparency, equity and recoverability, and falls with complexity and opacity. Appendix A writes this as a ratio for illustration, but the ratio form

has defects (A.1) and is not an estimable model, so no numerical coefficient is asserted and only the direction of each term is claimed.

A thesis that cannot fail is not worth stating. This one would be weakened by any of the following, each testable in participatory evaluation **[H]**:

- users given evidence bundles and reason codes explain conclusions, and identify what would change them, no better than users given conventional outputs;
- accessibility features raise users' confidence without raising the accuracy of their understanding;
- routes of contest exist but are not used, or do not change outcomes when used.

1.6 Research questions, scope and method

- **RQ1.** Can heterogeneous public telemetry be normalized without erasing source differences?
- **RQ2.** Do evidence gates reduce unjustified automation?
- **RQ3.** Can recoverability be measured distributionally?
- **RQ4.** Can disagreement between models and rules become useful audit information?

The work covers civic intelligibility, OSINT processing, market and infrastructure telemetry, evidence scoring, arbitration and governance analysis. It does not claim intelligence-agency capability or causal prediction of complex crises. Its outputs are triage indicators, not trading signals, investment recommendations or financial advice. These boundaries protect both users and academic credibility.

The method is design research informed by literature, author-supplied artifacts (Noffels, source archives) and comparative historical interpretation. The case studies generate requirements; the architecture operationalizes them; Chapter 4.5 defines the tests. No primary statistical dataset is analyzed in this text, and no empirical result about the prototype's performance is asserted in it.

1.7 Position relative to existing practice

Hannibal is not a new idea in every part, and it should be judged against what already exists.

The Admiralty (NATO) grading system. Intelligence reports have long been graded with a letter for the source's reliability (A to F) and, separately, a number for the information's credibility (1 to 6), so that a source's reputation does not silently contaminate an assessment of the item it supplies (NATO, 2003). Hannibal adopts the separation. It differs in that grades are machine-assigned from recorded properties rather than analyst-assigned; freshness and integrity are explicit dimensions; and contradiction is a first-class field instead of a downgrade of one digit. The Admiralty credibility grade already depends on corroboration by independent sources, which is exactly where Hannibal's hardest problem, source independence, arises (§3.2).

ICD 203. The U.S. Intelligence Community's analytic standards require, among other things, that products describe the quality and credibility of sources, express and explain uncertainty, distinguish underlying information from analysts' assumptions and judgments, incorporate analysis of alternatives, and explain changes in judgment (Office of the Director of National Intelligence, 2015). Hannibal turns several of these from writing guidelines into data-model constraints. It does not claim compliance with the directive, which governs a specific institutional setting. The directive's distinction between confidence and likelihood is worth keeping in view: Hannibal's scores are closer to confidence, the quality of support, than to the probability that a claim is true.

Forecast evaluation and calibration. Scoring rules such as the Brier score (Brier, 1950) and tournament-style evaluation of forecasters (Tetlock and Gardner, 2015) supply the vocabulary for asking whether stated confidence tracks outcomes. Guo et al. (2017) show that modern neural networks can be poorly calibrated, which is why this monograph refuses to treat any score as a probability until it has been calibrated.

Accountability and provenance. Raji et al. (2020) argue for end-to-end internal auditing across the lifecycle of AI systems, and the W3C PROV model (2013) offers a standard vocabulary for provenance in terms of entities, activities and agents. Hannibal's evidence records and audit ledger address the same concerns, and expressing them in PROV terms is an obvious interoperability step.

Regulation. For high-risk systems, the EU AI Act (Regulation (EU) 2024/1689) requires record-keeping (Article 12), transparency sufficient for deployers to interpret outputs (Article 13) and human oversight (Article 14). This monograph does not assert that Hannibal is a high-risk system under the Act; the articles are used as a public statement of comparable design goals.

1.8 Contributions and their evidential status

Table 1 · Contributions and the status of the support for each

#	Contribution	Status	Where
C1	Informational equity, the capacity to access, interpret and contest consequential evidence, as a dimension of human agency.	N, D	§1.4
C2	A comparative reading of Berlin and Belgium as sources of design requirements, with the provenance of each requirement shown.	S, interpretive	Part II
C3	An evidence-gated architecture in which provenance, freshness, contradiction, missingness and human review are first-class objects.	P	Part III
C4	Recoverability as a distributional, multi-dimensional record rather than a return-to-baseline flag.	D, P	§4.1
C5	Analysis of the prototype's own rules: saturation of the friction index, the cap on aggregate confidence, and defects of the ratio thesis.	R	§3.5, §3.6, A.1
C6	A validation design with falsification criteria and a reporting format.	H (design)	§4.5, §4.7, App. F

Status codes are defined in "How to read this monograph". C5 is checkable by hand from the equations in Appendix A.

PART II

Berlin and Belgium

2.1 Berlin as a natural experiment

Berlin's division and reunification changed access to surrounding economic density while many fixed geographic characteristics remained. Ahlfeldt, Redding, Sturm and Wolf (2015) use block-level land prices and employment for 1936, 1986 and 2006 to estimate localized production and residential externalities. Their evidence supports agglomeration mechanisms **[S]**. It does not, and was not designed to, adjudicate claims about political or economic systems.

Hannibal draws three narrower lessons, offered as interpretation of the case rather than as findings of the study. **First**, infrastructure is relational: the value of a location depends on the people, jobs and amenities reachable from it. Division changed the meaning of distance, so locations near the historic center could become peripheral when access was severed, and physical continuity can mask functional disconnection. **Second**, restoring a link is not restoring every flow. The fall of the Wall restored legal and physical mobility, but economic reorganization unfolded over time, and symbolic dates should not be used as binary recovery markers. **Third**, path dependence: housing, transport, firm location and expectations create persistence after a barrier is removed, so recoverability should include residual damage and the cost of re-coordination. Past structure constrains recovery but does not determine it.

2.2 What Berlin does not show

Land prices summarize expectations about accessibility, productivity, amenity and scarcity, and land value is not a complete measure of welfare or equity. Agglomeration can raise productivity through matching, sharing and learning, but density also creates congestion and distributional pressure, so dense connection can amplify both capability and disruption.

The distributional question, meaning who regained access, income, security and voice, and how quickly, is the one a recovery assessment most needs and the one this monograph cannot answer from the cited study. Aggregate redevelopment can coexist with displacement, unequal asset gains and divergent neighborhood trajectories, and claims about them require neighborhood-level data that are not estimated here.

For the architecture, the practical lesson is that a chokepoint matters because of the flows that depend on it. A system that merely counts disruption words is using text mentions as a proxy for network centrality, and the proxy is imperfect. The system must represent dependencies, alternatives and restoration states.

2.3 Belgium: three regions, one polity

Belgium offers a different pattern from Berlin: formal connection alongside divergent regional trajectories. Infrastructure remains open while industry, demography, politics and opportunity evolve differently. Belgium is not a controlled experiment and should not be reduced to language blocs.

Two sets of published figures make the caution concrete. First, employment rates differ sharply between regions (Table 2). Second, regional GDP per head hides large differences inside regions: the province of Walloon Brabant, in Wallonia, stands well above Flemish Brabant, Antwerp and Limburg, while Hainaut

and the Belgian province of Luxembourg, also in Wallonia, sit far below the EU average (Figure 1). A story told purely in terms of a Flemish–Walloon divide cannot account for that spread.

Table 2 · Employment rate, ages 20–64, 2023 (%)

Region	Employment rate	Gap to Flanders (points)
Flanders	76.8	—
Brussels	66.5	10.3
Wallonia	65.5	11.3

*Source: Federal Planning Bureau, BISA, Statistiek Vlaanderen and IWEPS (2024), regional economic outlook 2024–2029; EU2020 definition. Gaps are computed from the published rates **[S]**.*

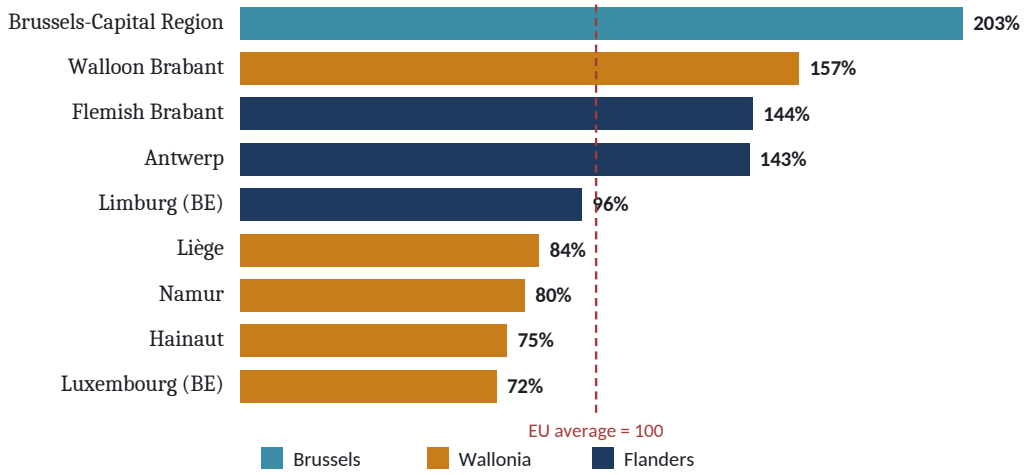


Figure 1 · GDP per head in purchasing power standards, 2018, as a percentage of the EU average, by Belgian province and the Brussels-Capital Region

*Sources: Eurostat (2020) for the Brussels-Capital Region; provincial figures as reported by The Brussels Times from the same Eurostat release **[S, secondary source for provinces]**. Provinces not reported in that source (East and West Flanders) are omitted, not estimated.*

2.4 Divergent adaptation

The standard historical account describes Wallonia as an early continental center of coal, steel and heavy industry, whose later deindustrialization was a structural shock to a region organized around concentrated sectors, and Flanders as having moved toward export manufacturing, services, ports and logistics, with Antwerp as a global–local node. This monograph does not test that account and treats it as background. Three cautions from it shape the design. Language does not cause prosperity or poverty; historical composition, network position, investment, education and access to trade matter. Recoverability after sectoral decline must track the rebuilding of ecosystems, meaning supplier networks, municipal revenue, identity and intergenerational expectations, and not only replacement job counts. And node centrality creates both capability and vulnerability: a port concentrates production and trade and also concentrates environmental, labor and geopolitical exposure.

Fiscal flows between regions are interpreted through competing narratives of solidarity, responsibility and autonomy. Hannibal's lesson is to separate measurable transfers from normative claims about justice, since numbers do not resolve political values by themselves.

2.5 Brussels: decision density and the commuter effect

Brussels is a constitutionally separate, multilingual and internationally connected region, and Belgium cannot be understood as a two-part comparison. EU institutions (European Union; European Commission), the Belgian government, NATO (NATO, n.d.), diplomacy, lobbying, media and civil society concentrate in a compact space. Its labor market and commuting zone cross regional boundaries: administrative borders and functional regions are not the same.

Decision density is the volume and reach of consequential decisions routed through a node **[D]**; a **governance chokepoint** is a node where rule-making, coordination or authorization concentrates, creating disproportionate downstream effects **[D]**. Brussels shows how centrality can arise from institutional connection rather than physical throughput. The terms are analytical and imply neither illegitimacy nor conspiracy. Decision density requires operational indicators before it can be measured (§4.2).

Informational distance. Citizens may experience regulations and policies far from the deliberations, evidence and trade-offs that produced them. Distance is cognitive and procedural as well as geographic, and accessible records and reasoned communication can reduce it. Complexity cannot be eliminated without losing necessary detail.

The place-of-work problem. In 2023 the Brussels-Capital Region's GDP per head reached 190.6% of the EU average in purchasing power standards, fifth among EU regions at the NUTS 2 level, and Eurostat itself attributes the high values of regions such as Brussels partly to a large inflow of commuting workers (Eurostat, 2025) **[S]**. The same year's employment rate among Brussels residents aged 20–64 was 66.5%, ten points below Flanders (Table 2). These figures are consistent with a distinction the monograph relies on, between output generated at a place of work and the welfare of the people who live there, though they do not by themselves establish it: employment rates are not poverty rates. Any indicator in Hannibal must therefore carry its scope, place-of-work or resident population, as a field.

2.6 Language, plurality and cohesion

French in Brussels, Dutch in Flanders and French in Wallonia do not mechanically determine prosperity. Language interacts with labor markets, education, administration, mobility and networks, and economic trajectories arise from institutions and history. Cultural explanations should not replace structural evidence; the provincial spread in Figure 1 points the same way.

Belgium's German-speaking community reminds the analysis that plurality exceeds a binary map. Small communities can hold institutional significance disproportionate to their population and deserve visibility in equity analysis. Inclusion should not be symbolic: data and interfaces must support minority-language access.

Finally, Belgium shows that a polity can persist through negotiated difference rather than complete convergence. Hannibal's arbitration philosophy mirrors this, since disagreement can be preserved while action remains possible. Political compromise and algorithmic arbitration are not equivalent, and the analogy is limited.

2.7 From cases to requirements

Berlin represents severed and restored connection; Wallonia and Flanders divergent adaptation; Brussels concentrated governance. Together they show that connectivity, cohesion and equity are distinct dimensions, and no single metric should collapse all three. Table 3 states what each case contributes to the design so that the link from Part II to Part III can be inspected.

Table 3 · Traceability from cases to design requirements

Req.	Source in Part II	Requirement	Implemented in
R1	Berlin: infrastructure is relational; a chokepoint matters through dependent flows	Represent dependencies, alternatives and restoration states, not disruption-word counts.	§3.6 (multiplier), §4.2
R2	Berlin: restoring a link is not restoring every flow; path dependence	Track restoration in stages and record residual damage and re-coordination cost.	§4.1
R3	Berlin: who benefits; Brussels: place-of-work output vs resident welfare	Every indicator carries its scope; recovery is reported by population group.	§3.3, §4.1
R4	Brussels: decision density	Governance-node scores must not rest on institution count alone.	§4.2
R5	Belgium: plurality; German-speaking community	Multilingual and minority-language access is an evaluation criterion, not an add-on.	§4.3
R6	Belgium: cohesion without uniformity	Preserve disagreement between analytical streams while action remains possible.	§3.8

The cases generate these requirements; they do not validate the software. Only the Berlin case rests on an identified empirical study; the Belgian material is descriptive.

PART III

Technical Architecture

3.1 Design principles

Hannibal separates collection, normalization, distillation, evidence assessment, arbitration, gating, presentation and audit (Figure 2). The purpose is to prevent a raw feed from becoming an answer without intermediate controls. The separation increases engineering complexity, and whether it is worth that cost is an empirical question (§4.5).

> *The analytical discipline proposed here is to separate observation, inference, evaluation and action. Each stage must remain identifiable in the record. A fluent explanation is not evidence; an institutional source is not infallible; agreement is not proof; and missing data are not zero.*
>
> *For Project Hannibal, the practical implication is that the interface should help a reader ask better questions. It should identify what is known, what is inferred, what is contradicted, what is stale and what requires a person to decide.*

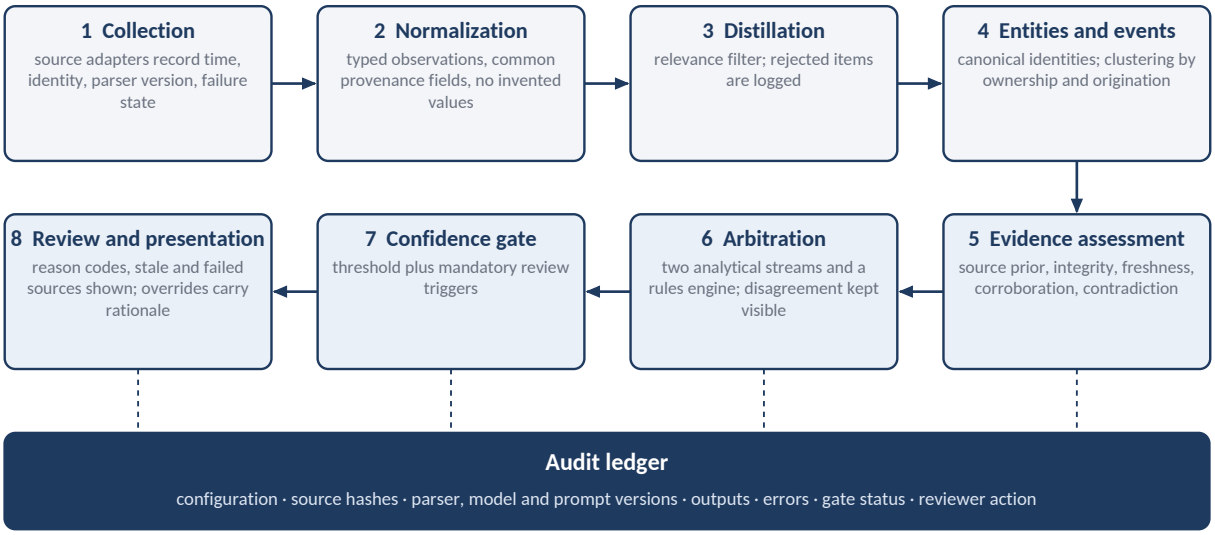


Figure 2 · The Hannibal evidence pipeline. Every stage writes to the audit ledger.

3.2 From feeds to events

Market, news, weather (National Weather Service), aviation, regulatory, prediction-market and subsea sources enter through source-specific adapters. Table 4 states, for each stage, what it does and what it must not do.

Table 4 · Pipeline stages from collection to event

Stage	What it does	What it must not do
Collection	Adapters record retrieval time, source identity, parser version and failure state.	Treat public availability as reliability, or as permission for every reuse.
Normalization	Turns heterogeneous payloads into typed observations with common provenance fields, keeping references to original content.	Invent absent values. A normalized record is not verified evidence.
Distillation	Prioritizes candidate events by recency, domain, location and potential dependency impact; keeps a log of rejected items.	Hide what was filtered out. Keyword filters can reproduce blind spots and duplication.
Entity resolution	Assigns canonical identities with confidence levels to names, organizations, locations and events.	Force ambiguous entities into one identity, or let one event pose as many.
Event clustering	Groups near-duplicate and syndicated reports into event candidates; counts independence by ownership and origination.	Count URLs as independent sources. Perfect independence is often unknowable.

3.3 The evidence record

An evidence record holds an identifier, source class, timestamps, content hash, integrity state, freshness, corroboration, contradiction and scope (Appendix B). Records are immutable; corrections create superseding records. Hashes prove byte continuity only within an appropriate trust design. The scope field carries requirement R3: an observation about a place of work is not an observation about the people who live there.

3.4 Scoring evidence

Source class provides a prior. Table 5 lists the classes and states, for each, what it can establish and its principal failure mode. Weights attached to classes reflect a policy prior **[P]** that must be calibrated by domain, versioned and published; institutional status is not a truth guarantee.

Table 5 · Source classes

Class	What it can establish	Principal failure mode
Signed records	Origin and content integrity, cryptographically.	A signature says nothing about whether the content is true.
Transaction records	That a transaction was recorded, as in SEC Form 4 filings.	Inferring motive from a filing alone.
Provider alerts	The provider's own operational status.	Not independent of the provider; may lag or understate.
Official publications	An institution's stated position or decision.	Treating institutional status as proof of accuracy.
Identified testimony	One named person's account.	Perspective, interest and memory.
Anonymous reports	That a claim exists; a lead.	Cannot be attributed or held to account.
Unverified claims	That the claim was made.	Repetition mistaken for corroboration.

Four further dimensions modify the prior. **Integrity** captures whether origin and content have been authenticated or changed; signatures, transport security and independent hashes contribute different protections, and integrity is not factual accuracy. **Freshness** declines as evidence ages relative to domain-specific validity windows; weather, market prices and legal filings need different decay functions, and old evidence can remain historically valid while operationally stale. **Corroboration** rewards

independent confirmation of the same material claim; the architecture stores which claim is corroborated, not only a count, because correlated sources can create false confidence. **Contradiction** is represented explicitly and weighted by source quality, recency and claim scope; high-quality conflict triggers review even when an aggregate score remains high, and contradiction detection by language models requires human validation.

An individual policy score combines these and is clipped to a bounded range. It is an operational triage value, not a probability unless calibrated, and its weights must be versioned and published.

3.5 Aggregate confidence

A bundle of evidence is scored by combining mean item quality with the weakest critical item, so that a long chain of strong records cannot hide one unsupported premise **[P]**:

$$Q = \text{clip}[\ 0.75\ (\text{mean item score}) + 0.25\ (\text{minimum critical-item score})\ ]$$

The minimum term helps, but not as much as its purpose implies **[R]**. Assume item scores lie in [0, 1]. The weights sum to 1, so Q also lies in [0, 1] and the clip is only a guard. Because the mean cannot exceed 1, however:

$$Q \leq 0.75 + 0.25\ s_{\min}$$

where $s_{\min}$ is the score of the weakest critical item. An entirely unsupported critical premise ($s_{\min} = 0$) still allows Q to approach 0.75 when the other items are strong. Table 6 gives worked cases; these are arithmetic illustrations, not empirical results.

Table 6 · Aggregate confidence: worked cases (illustrative)

Case	Mean	Min. critical	Q
Nine items at 0.90 and one critical item at 0.10	0.820	0.10	0.640
Nine items at 0.90 and one critical item at 0.00	0.810	0.00	0.607
Ninety-nine items at 1.00 and one critical item at 0.00	0.990	0.00	0.742
Nine items at 1.00 and one critical item at 0.50	0.950	0.50	0.837
Ten items at 0.90, all critical	0.900	0.90	0.900

The implication is a design requirement. If the gate threshold τ is at or below 0.75, an unsupported critical premise can pass on the aggregate score alone, so the gate needs a separate floor on the weakest critical item. The mandatory-review triggers in §3.8 partly supply this (missing provenance), but a floor on $s_{\min}$ should be added explicitly. Separately, the method can over-penalize non-critical weak items unless claim dependencies are modeled.

3.6 Composite friction and the chokepoint multiplier

The prototype friction index combines kinetic, infrastructure, labor, aviation, weather and prediction-market components into a base score, then scales it by a chokepoint multiplier and clips at 1. Its purpose is triage of possible systemic stress, not causal explanation, and the hand-selected weights and clipping prevent any claim of empirical validation **[P]**.

$$F_t = \min\{ C_t B_t, 1 \}, C_t = 1 + 0.15\, h_t$$

$$B_t = 0.20\, K_t + 0.20\, I_t + 0.10\, L_t + 0.10\, A_t + 0.10\, W_t + 0.30\, P_t$$

Table 7 · Friction variables

Symbol	Component	Weight	Note
K	Kinetic	0.20	
I	Infrastructure	0.20	
L	Labor	0.10	
A	Aviation	0.10	
W	Weather	0.10	
P	Prediction-market momentum	0.30	Largest weight; see the objection in §4.7
h	Chokepoint measure	0.15 (in *C*)	Keyword-hit counts in the prototype; a demonstration proxy

*Weights sum to 1.00 by construction and are hand-selected **[P]**. Components are assumed to lie in [0, 1], as is h; the equation as stated does not say so.*

Saturation [R]. Under those assumptions *B* lies in [0, 1] and *C* in [1, 1.15]. Then $F = 1$ whenever $B \geq 1/C$: at $h = 1$ that is any *B* of 0.870 or more, and at $h = 0.5$ any *B* of 0.930 or more (Figure 3, Table 8). Among nodes with very high base stress the multiplier therefore cannot separate a high-dependency node from a low-dependency one, which is precisely where triage most needs it to. Elsewhere the multiplier adds at most 0.15 × *B*, a rescaling of at most 15%.

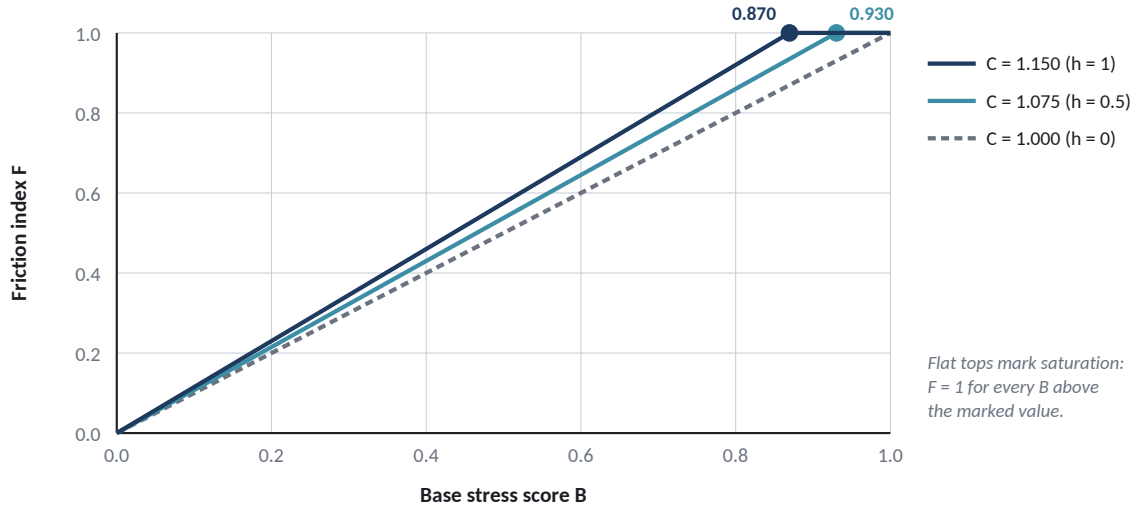


Figure 3 · Friction index F against base stress B for three values of the chokepoint measure h

Table 8 · Friction index F for selected base scores (illustrative)

Base score B	$h = 0$	$h = 0.5$	$h = 1$
0.40	0.400	0.430	0.460
0.60	0.600	0.645	0.690
0.80	0.800	0.860	0.920
0.90	0.900	0.968	1.000
0.95	0.950	1.000	1.000

Recommendation. Report B_t and h_t as separate channels and use F_t only as a sort key, or replace the hard clip with a monotone saturating map so that rank order survives at the top. Whichever is chosen should be evaluated against the weight-sensitivity test in §4.5. A mature multiplier should use flow share, substitutability, duration and repair time, in line with the Berlin lesson that a chokepoint matters through the flows that depend on it (R1).

3.7 Signals and their limits

Table 9 · Signals used as inputs and what they cannot establish

Signal	What it observes	What it cannot establish
Prediction-market momentum	Rapid changes in implied probabilities, read as shifts in collective expectation. Direction, liquidity, market rules and participant composition stay visible.	Ground truth. Market prices are not facts about the world.
Subsea and ocean telemetry (DART)	Environmental observations, retained with station, timestamp, measurement quality and domain interpretation (NOAA National Data Buoy Center).	Cause or threat. A sensor anomaly is not an explanation.
Regulatory filings (SEC Form 4)	Structured reports of insider transactions, represented neutrally (U.S. Securities and Exchange Commission).	Motive. Parsing and issuer identity need robust test fixtures.
Market telemetry	Prices, returns and volumes, with timestamp, venue and data delay known.	A unique cause for a price movement.

3.8 Arbitration, the confidence gate and human review

Two analytical streams and one rules engine produce separate structured outputs. Disagreement, failure and latency remain visible instead of being silently merged (R6). Three agreeing engines can still be wrong, especially when they share inputs, so agreement is reported and not treated as proof.

The gate (Figure 4) blocks action when support falls below the threshold τ or when a trigger condition applies. Mandatory review follows high-quality contradiction, missing provenance, novel event type or irreversible consequence, and the ethical triggers in §4.4. Thresholds need calibration against labeled cases **[P]**.

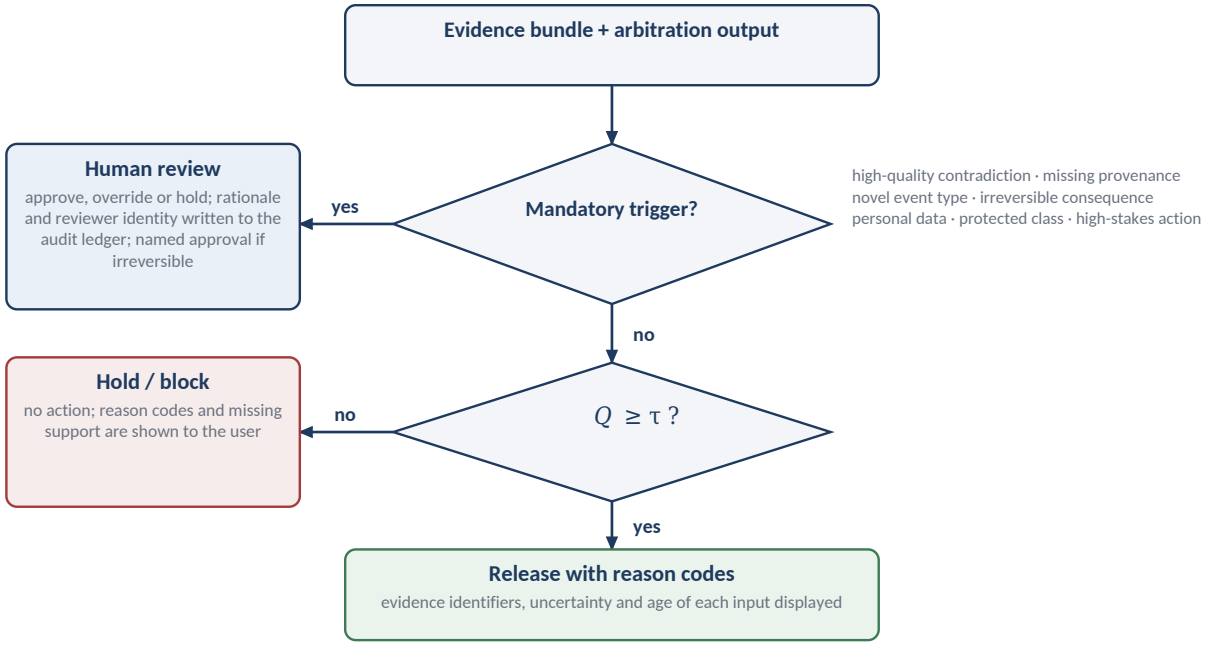


Figure 4 · The confidence gate. Mandatory triggers are checked before the score threshold.

Reviewers receive evidence bundles, reason codes, uncertainties and competing interpretations. Overrides require a rationale and generate auditable records with reviewer identity. Human review can introduce delay, fatigue and bias, and reviewers may defer to the machine's recommendation without scrutiny, so override rate and review time are measured, not assumed (§4.5).

3.9 The language model and the interface

A local language model synthesizes and explains after evidence processing. Prompts require evidence identifiers and separate observation from inference. The model is not a source of truth, and its output is never stored as evidence (Appendix C, test 5). Whether its explanations improve comprehension over templated reason codes is a hypothesis to be tested, and if they do not, the model should be removed **[H]**.

The presentation layer offers sector views, maps, timelines and confidence panels. Interfaces must disclose stale data, source failure and uncertainty prominently, and visual polish must not imply precision.

3.10 Audit, security and failure semantics

Each run records configuration, source hashes, parser versions, model versions, prompts, outputs, errors, gate status and reviewer action. Audit completeness supports correction and reproducibility, while sensitive records need retention limits and access controls. The audit vocabulary maps naturally onto the W3C PROV model (2013).

Encryption at rest, password-based key derivation and least-privilege design reduce casual disclosure. Per-installation salts, protected secrets and host security are necessary for stronger assurance (National Institute of Standards and Technology), and a compromised running host can expose plaintext.

Failure has to be distinguishable from absence. Table 10 lists the states that must stay distinct. Error transparency may reduce apparent uptime but increases epistemic reliability. Fast and slow polling paths

should expose last success, retry budget, queue depth and dependency health, and single-flight controls prevent overlapping inference work. Reliability claims require service-level measures.

Table 10 · Failure states and how they must not be displayed

State	Meaning	Must not be rendered as
Unavailable	The source could not be reached.	Zero, or “no events”.
Empty	The source responded and returned no items.	Verified absence.
Stale	The latest data are older than the domain’s validity window.	Current.
Contradicted	Credible sources disagree.	An averaged value.
Verified absence	Positive evidence that nothing occurred within the stated scope.	Unavailable.

3.11 Architecture synthesis

The architecture expresses the thesis as controls: provenance for evidence, explanation for transparency, accessibility for equity and restoration states for recoverability. Technology serves the civic argument instead of replacing it, and validation determines whether the design actually achieves those aims.

PART IV

Recoverability, Governance and Validation

4.1 Recoverability defined

Recoverability is the capacity and speed with which useful, legitimate and equitably distributed functions are restored after disruption **[D]**. It differs from resistance, from resilience in Holling's (1973) sense of a system's capacity to persist through change, and from simple return to baseline, because baseline conditions may themselves be unjust. The dimensions in Table 11 are reported separately; no single score is offered.

Table 11 · Recoverability dimensions

Dimension	What is examined	Typical pitfall
Accessibility	Restored access to services, mobility, information and institutional remedies, measured across affected groups.	Nominal availability hides cost or language barriers.
Time	Time to partial and full restoration, for different users' different recovery clocks.	A single average conceals long tails.
Substitution capacity	Alternative routes, suppliers, institutions and channels, including quality, affordability and time to activation.	Nominal alternatives may be unusable in practice.
Cohesion	Whether trust, cooperation and legitimacy make restored infrastructure socially useful.	Trust is not obedience; indicators need cautious reading.
Distribution	Whether benefits concentrate and burdens persist among the least powerful; who regained agency and who remained exposed.	Fairness norms are chosen, and must be stated.
Residual damage	Debt, trauma, deferred maintenance or institutional distrust carried by systems that function.	Some harms cannot be reduced to one score.

Distributional judgment requires a stated norm. Rawls's (1999) difference principle, which asks whether inequalities benefit the least advantaged, is one candidate; this monograph does not adopt it, but insists that whatever norm is used be declared. A recoverability record stores the affected function, population, baseline, restored fraction, time, substitutes, residual damage, confidence and evidence links (Appendix B), which keeps the "for whom" question inside the data model. Population categories require privacy and anti-discrimination safeguards.

4.2 Governance nodes

Decision density can be explored through institution count, rule reach, dependency, participation access, transparency and appeal paths. A governance-chokepoint score must never be built from institution count alone (R4). The framework remains a research proposal **[H]**.

4.3 Equity evaluation

Equity asks whether access to evidence, explanation and remedy is fairly distributed. User testing should include language, disability, expertise, time and socioeconomic constraints, and universal design requires participation, not assumptions. Three measures are proposed **[H]**: comprehension accuracy (can a user state what a conclusion rests on and what would change it), time to understanding, and the rate and outcome of contested decisions across groups. These are also the tests that could weaken the thesis (§1.5).

4.4 Ethical boundaries

The system must not infer live personal location, attribute attacks from metadata alone or convert weak OSINT into accusation. Protected classes, personal data and high-stakes actions trigger mandatory safeguards, and ethical constraints need enforceable tests (Appendix C, tests 8 and 9). Its outputs are triage indicators and are not to be presented as trading signals, investment advice or accusations.

Historical awareness supports refusal of dehumanizing classification, opaque authority and "just following procedure" reasoning. Documentation should preserve responsibility instead of diffusing it across components: every consequential output has a named human owner. As stated in §1.2, this implies no equivalence between AI errors and genocide.

4.5 Validation design

This chapter specifies what a validation result must look like and asserts no result. Historical fixtures, labeled event clusters, calibration sets, ablations and human baselines test the architecture, and performance includes precision, recall, calibration, latency, review burden and correction rate. Anecdotes are not validation.

Ground truth. Let $y_i = 1$ if claim i is later confirmed and 0 otherwise, under adjudication rules fixed before scoring. A second rater labels a sample, and agreement is reported (Cohen, 1960). **Calibration.** Because Q is a triage value, not a probability, the tests ask how well it behaves as one:

$$\text{Brier} = (1/n)\,\Sigma\,(Q_i - y_i)^2,\ \text{ECE} = \Sigma_b\,(n_b/n)\,|\,\text{acc}(b) - \text{conf}(b)\,|$$

Brier (1950) is the mean squared error of a stated confidence; ECE compares observed correctness with mean confidence within bins of Q. **Selective performance.** At each threshold τ the gate releases a fraction of claims (coverage), and the error rate among those released is the risk. The risk–coverage curve, compared with the ungated baseline, is the direct test of whether the gate reduces unjustified automation.

Table 12 · Estimands, baselines and what would count against each claim

Claim	Measure	Baseline	Counts against the claim if
Scores carry information	Brier, ECE, reliability table, AUROC of Q against y	Constant prediction; single-engine score	Q is no better than a constant, or is badly miscalibrated with no monotone relation to y.
The gate reduces unjustified automation	Risk–coverage curve; review rate	No gate; single engine	At matched coverage, gated risk is not lower than ungated risk.
Arbitration exposes useful disagreement	Error rate when engines disagree vs agree	Any single engine	Disagreement is unrelated to error.
Source independence is captured	Agreement between cluster-based counts and labeled independence	URL count	Cluster counts do no better than URL counts.
Human review adds value	Override rate; errors caught vs introduced; review time	Automatic release	Override rate is near zero with reviewers shown disagreement data, or overrides do not reduce error.
Weights are not decorative	Rank stability of F under weight perturbation (Kendall's ⊠)	Equal weights	Rank order flips under plausible reweighting.
Each component is needed	Change in Brier and risk when a component is removed	Full system	Removing a component leaves results unchanged.

Safeguards. Scoring at time t uses only data available at t. Fixtures are frozen and hashed before evaluation. Thresholds are set on a calibration split and evaluated on held-out data. Intervals are bootstrapped, negative results are reported, and the reporting format in Appendix F is fixed in advance so that only aggregate tables, not record-level data, are needed to assess the claims.

4.6 Red teaming and governance of Hannibal

Adversarial tests include duplicate-source floods, forged provenance, stale-but-plausible data, prompt injection and coordinated narratives. The system should fail visibly and safely, and red-team findings should produce versioned mitigations.

A deployment requires a permitted-use policy, model and weight change control, reviewer roles, incident reporting, correction and appeal. Independent review is desirable for consequential uses, and governance must scale with risk. Ostrom's (1990) design principles for long-enduring institutions, which include monitoring and accessible conflict-resolution mechanisms, are a useful check on whether these arrangements can persist.

4.7 Objections and replies

The strongest objections are stated here in their strongest form, each with the reply and the result that would vindicate the objector.

1. Objection. *The weights are arbitrary, so the scores are decoration.*

Reply. Partly right: source priors and friction weights are hand-selected **[P]**, and the largest single friction weight (0.30) falls on prediction-market momentum, a signal the monograph itself says is not ground truth. The claim is limited to triage, with weights versioned and published, and the sensitivity and calibration tests in §4.5 apply. If rank order flips under plausible reweighting, or if Q is uncorrelated with adjudicated correctness, the scores add nothing and should be dropped.

2. Objection. *Publishing the gate lets adversaries game it.*

Reply. Transparency has a security cost. The commitment is to transparency of the decision path for those affected, not to publication of every exploitable threshold; thresholds may be disclosed to auditors and withheld publicly. That is a limit on the equity claim and is stated where it applies. The red-team suite tests forged provenance and duplicate floods.

3. Objection. *Informational equity is too vague to measure.*

Reply. Today it is a set of dimensions **[D]**, not an index. The comprehension, time and contest measures in §4.3 make it empirically contestable. If no measure discriminates between designs, the concept should be abandoned as an evaluation tool.

4. Objection. *Two cases are anecdotes and cannot justify design requirements.*

Reply. They cannot validate software, and are used only to generate requirements (Table 3), each independently testable. The weakness is real: one case rests on an identified study and the other is descriptive.

5. Objection. *Human review launders automation; the reviewer becomes a rubber stamp.*

Reply. Automation bias is a real risk. Override rate, review time and errors caught versus introduced are measured; irreversible actions need named approval. Near-zero overrides with reviewers who can see disagreement data would show the control is not working.

6. Objection. *A local language model adds fluency, not evidence.*

Reply. Agreed, which is why it is confined to post-evidence explanation, cannot create evidence records, and is removable. An ablation against templated reason codes tests whether it earns its place.

7. Objection. *Corroboration is meaningless if independence cannot be established.*

Reply. Perfect independence is unknowable. The system clusters by ownership and origination, stores which claim is corroborated, and states residual correlation as a limitation. A labeled independence sample tests whether clustering beats URL counts.

8. Objection. *The thesis equation is unfalsifiable.*

Reply. The ratio form is not an estimable model and has defects (A.1). The directional claims are the thesis, and §1.5 states what would falsify them.

4.8 Limitations and research agenda

The prototype depends on changing public endpoints, hand-selected weights, uncertain source independence and local-model behavior. The comparative cases ground concepts but do not validate software performance. Three properties of the current rules limit what can be concluded from them: the friction index saturates, the aggregate confidence score has a cap, and the ratio thesis is defective (Appendix A.1). Public release should state these limits prominently.

Future work includes measured network graphs, distributional recovery data, multilingual evaluation, calibration, governance-node analysis and participatory design. The test of the program is whether users gain accurate, contestable understanding. Feature count is not research progress.

Afterword

Project Hannibal remains unfinished because the problem it addresses is unfinished. Systems will continue to grow more complex, and institutions will continue to face the temptation to substitute procedural confidence for explanation.

The project's next measure of success is not the number of feeds, models or screens it can add. It is whether a reader can understand the evidence, recognize uncertainty, challenge a conclusion and see who benefits from recovery.

If Hannibal becomes useful, it will be because it keeps technical ambition subordinate to human dignity. Evidence. Transparency. Equity. Recoverability. Human agency.

Appendices

Appendix A · Core equations

NORMATIVE THESIS (ILLUSTRATIVE)

$$\text{Agency} \propto \frac{\text{Evidence} \times \text{Transparency} \times \text{Equity} \times \text{Recoverability}}{\text{Complexity} \times \text{Opacity}}$$

*This is a conceptual relation, not an estimated law **[N]**.*

PROTOTYPE FRICTION

$$F_t = \min\{\, C_t \,[\, 0.20\, K_t + 0.20\, I_t + 0.10\, L_t + 0.10\, A_t + 0.10\, W_t + 0.30\, P_t \,],\, 1 \,\}$$

CHOKEPOINT MULTIPLIER

$$C_t = 1 + 0.15\, h_t$$

AGGREGATE EVIDENCE CONFIDENCE

$$Q = \text{clip}[\ 0.75\ (\text{mean item score}) + 0.25\ (\text{minimum critical-item score})\ ]$$

*All coefficients are prototype policy choices and require calibration before operational use **[P]**.*

A.1 Properties and defects

Thesis ratio. Three defects follow from the form **[R]**. (i) Transparency appears in the numerator and opacity in the denominator, but they are close to opposites, so the relation counts one underlying property twice. (ii) As opacity approaches zero the ratio grows without bound, implying unbounded agency from a fully transparent system, which is not a claim anyone should make. (iii) The product form makes every term essential, so a zero in any numerator term sets agency to zero regardless of the others; that may be intended, but it is a strong substantive commitment. For these reasons the monograph asserts only the direction of each term.

Aggregate confidence. With scores in [0, 1], Q lies in [0, 1] and $Q \le 0.75 + 0.25\, s_{\min}$ (§3.5). **Friction.** With components and h in [0, 1], F saturates at 1 when $B \ge 1/(1 + 0.15\, h)$, which is 0.870 at $h = 1$ (§3.6). Both results follow directly from the stated formulas and the stated assumptions.

Appendix B · Data schemas

Evidence record

`evidence_id`	`claim_id`	`source_class`
`source_uri`	`published_at`	`retrieved_at`
`parser_version`	`content_hash`	`integrity_state`
`freshness`	`corroborates[]`	`contradicts[]`
`scope`	`confidence`	`supersedes`

Arbitration record

```
run_id                  evidence_ids[]          engine_a
engine_b                rules_engine            disagreements[]
successful_engines      recommendation          gate_state
reason_codes[]
```

Recoverability record

```
event_id                affected_function       affected_population
baseline                restored_fraction       restoration_time
substitutes[]           cohesion_notes          distribution_notes
residual_damage         confidence              evidence_ids[]
```

Audit record

```
actor                   utc_time                action
configuration_hash      model_version           prompt_version
before_hash             after_hash              reviewer
rationale
```

Appendix C · Acceptance tests

Each requirement is stated as a prohibition that a test can violate. The verification column proposes a way to test it against the architecture described in Part III **[P]**.

Table 13 · Acceptance tests and proposed verification

#	Requirement	Proposed verification
1	No claim without an evidence identifier.	Schema constraint: claim rows require non-empty evidence identifiers; property test over generated records.
2	No missing value rendered as zero.	Rendering test: unavailable, empty and verified absence map to distinct labels (Table 10).
3	No stale observation shown without age.	Interface test: every observation displays its age, with a stale flag beyond its validity window.
4	No high-quality contradiction silently averaged away.	Fixture with two high-quality conflicting sources must produce a review flag, not a mean.
5	No model text treated as source evidence.	Evidence store rejects records whose origin is model output; model text carries a generated-by label.
6	No consequential action below the confidence threshold.	Gate test at threshold boundaries (just below, at, just above).
7	No irreversible action without named human approval.	Action interface rejects requests lacking a reviewer identity.
8	No personal-location inference.	Red-team fixtures requesting an individual's location must be refused and logged.
9	No attribution from metadata alone.	Fixtures containing only metadata indicators must not produce an attribution output.
10	Every override carries a reason code and reviewer identity.	Audit-record schema requires both fields; writes without them are rejected.

Appendix D · Author and CV note

The narrative presents Emmanuel “Bram” Noffels as a Belgian author shaped by multilingual environments, cross-border work, support and investigative operations, and continuing volunteer association with Bouckenborgh Park. Employment and volunteer affiliations are author-supplied and may be verified directly with the relevant organizations under their policies (Noffels, curriculum vitae).

Appendix E · Ledger of central claims

Each central claim is listed with its status, its support and the result that would weaken it. Codes are defined in “How to read this monograph”.

Table 14 · Ledger of central claims

ID	Claim	Status	Support	Would be weakened by
L1	Informational equity is a dimension of human agency.	N, D	Conceptual basis in Sen (1999); no empirical measure yet.	Comprehension studies show no difference across designs.
L2	Berlin's division and reunification changed access to economic density.	S	Ahlfeldt et al. (2015).	—
L3	Restoring a link does not restore every flow; path dependence follows.	Interp.	Interpretation of the Berlin case, not a stated finding of the study.	Evidence of rapid, complete adjustment.
L4	Brussels output per head is far above the EU average, partly through commuting.	S	Eurostat (2025): 190.6% in 2023; commuting inflow cited.	—
L5	Employment rates differ by about 10 points between Flanders and Brussels or Wallonia.	S	Federal Planning Bureau et al. (2024).	—
L6	Provincial spread is inconsistent with a purely language-bloc explanation.	S	Eurostat 2018 data via The Brussels Times (secondary).	Primary NUTS 3 table showing the opposite.
L7	Wallonia's and Flanders' historical trajectories.	Backgr.	Background only; not evidenced in this monograph.	—
L8	The architecture separates observation, inference, evaluation and action.	P	Design; tested by acceptance tests 1–10.	Any test failing on the prototype.
L9	$Q \le 0.75 + 0.25\, s_{\min}$.	R	Follows from the definition, scores in $[0, 1]$.	A different definition of Q.
L10	F saturates when $B \ge 1/C$.	R	Follows from the definition, components and h in $[0, 1]$.	A different range for h.
L11	The ratio form of the thesis is defective as a model.	R	Appendix A.1.	—
L12	Source priors and friction weights are policy choices.	P	Hand-selected; calibration required.	Calibration supports them.
L13	The gate reduces unjustified automation.	H	Untested in this text.	No gain at matched coverage (§4.5).
L14	Three-way arbitration exposes useful disagreement.	H	Untested in this text.	Disagreement unrelated to error.
L15	Local-model explanations improve comprehension over reason codes.	H	Untested in this text.	No comprehension gain.
L16	Recoverability can be measured distributionally.	H	Untested in this text.	No stable measure emerges.

Appendix F · Reporting format for validation results

Results are to be reported as the aggregate tables below. No record-level data are needed to assess the claims in Table 12.

Table 15 · Aggregate tables for a validation report

Table	Contents
R1 Calibration	For each bin of Q: number of claims, mean score, observed correctness rate.
R2 Headline metrics	Number of claims, prevalence of correct claims, Brier, ECE, AUROC, each with a 95% bootstrap interval.
R3 Risk–coverage	For each threshold: coverage, risk among released claims, share routed to human review.
R4 Review	Review rate, override rate, median review time, corrections after release.
R5 Latency	Median, 90th and 99th percentile end-to-end latency.
R6 Subgroups	Number of claims, Brier and ECE by source class, domain and language.
R7 Weight sensitivity	Distribution of Kendall's ⊠ between the baseline ranking of F and rankings under perturbed weights.
R8 Ablations	Change in Brier and risk when each component is removed.

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